

Structural Boundaries and Performance Limitations of STL Decomposition Preprocessing in Multi-Step Meteorological Time Series Forecasting

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Abstract—This study investigates the structural boundary conditions and empirical efficacy of Seasonal-Trend decomposition using Loess (STL) when deployed as an explicit preprocessing step for multi-step time series forecasting. Utilizing high-frequency daily mean temperature data from Hanoi, Vietnam (2015–2025, $n = 3,742$), we conduct a rigorous symmetrical evaluation between three end-to-end models (SARIMA, XGBoost, LSTM) and three decompose-then-forecast hybrid pipelines (STL+SARIMA, STL+XGBoost, STL+LSTM) across horizons $t + 1$, $t + 3$, and $t + 7$. Counter to conventional theoretical assumptions, empirical results demonstrate severe performance degradation when utilizing STL decomposition. At a short-term horizon ($t + 1$), the STL+LSTM pipeline yields a Root Mean Squared Error (RMSE) of 2.8482°C , representing an 84.35% increase in error compared to the end-to-end LSTM model (RMSE = 1.5450°C). Symmetrical degradation is verified across all machine learning and deep learning configurations via the Diebold-Mariano test ($p < 0.01$). Diagnostic post-mortem analysis reveals that the core breakdown is driven by the structural properties of the series: an exceptionally low strength of trend ($F_T = 0.0140$) coupled with high seasonal dominance ($F_S = 0.8016$). Under these conditions, the Loess-extracted trend component injects additive noise rather than predictive signal, while restricting non-linear models to model non-stationary residuals induces critical overfitting. We propose a conditional framework establishing quantitative boundaries for decomposition-based time series pipelines.

Index Terms—Time series forecasting, STL decomposition, Long Short-Term Memory, XGBoost, Strength of Trend, Multi-horizon forecasting, Diebold-Mariano test.

I. INTRODUCTION

Multi-step daily temperature forecasting is essential for grid load balancing, agricultural scheduling, and metropolitan climate resilience planning. Due to the complex, non-linear dynamics governing atmospheric interactions, standard time series methodologies frequently struggle with long-term dependencies and high-frequency variations. A widely adopted paradigm in contemporary forecasting literature is the *decompose-then-forecast* approach, where raw data is systematically separated into isolated structural components prior to optimization.

Seasonal-Trend decomposition using Loess (STL) represents the state-of-the-art in non-parametric decomposition,

prized for its robustness against extreme outliers and capability to handle highly non-linear seasonal variations. The fundamental assumption underlying hybrid frameworks is that filtering out deterministic components reduces the mathematical complexity of the data stream, enabling downstream statistical or machine learning architectures to optimize solely on simplified stationary sub-signals. Large-scale forecasting benchmarks, such as the M4 Competition [10], have continuously underscored the potential of hybrid systems, yet a formal characterization of the structural conditions under which decomposition impairs forecasting remains an open research gap.

This paper addresses this gap by utilizing a decade-long daily meteorological dataset from Hanoi, Vietnam (2015–2025). We implement a strict, symmetrical experimental design comparing end-to-end architectures against their STL-preprocessed equivalents.

Our core contributions are threefold: 1) providing clear empirical proof that forcing STL decomposition on an asset with an exceptionally weak long-term trend systematically degrades multi-step accuracy; 2) validating the statistical significance of this degradation across statistical, machine learning, and deep learning paradigms using the Diebold-Mariano framework; and 3) defining quantitative structural boundaries based on component variance metrics to serve as an architectural guardrail for time series engineering.

II. MATHEMATICAL FRAMEWORK AND METHODOLOGY

A. STL Decomposition and Feature Quantification

The additive STL framework isolates the observed time-series target Y_t into trend (T_t), seasonal (S_t), and remainder (R_t) components:

$$Y_t = T_t + S_t + R_t \quad (1)$$

The decomposition is executed via a system of two nested loops. The inner loop updates the seasonal and trend components through alternating steps of subseries smoothing and low-pass filtering utilizing localized polynomial regression (Loess) [3]. The outer loop computes robustness weights ρ_t

based on a bisquare function of the remainder R_t to reduce the influence of anomaly spikes:

$$\rho_t = B \left(\frac{|R_t|}{6 \cdot \text{median}(|R_t|)} \right) \quad (2)$$

To mathematically define the information profile of the target series, we adopt the metric proposed by Wang et al. [11] to quantify the strength of trend (F_T) and strength of seasonality (F_S):

$$F_T = \max \left(0, 1 - \frac{\text{Var}(R_t)}{\text{Var}(T_t + R_t)} \right) \quad (3)$$

$$F_S = \max \left(0, 1 - \frac{\text{Var}(R_t)}{\text{Var}(S_t + R_t)} \right) \quad (4)$$

B. Predictive Architectures

The experimental matrix comprises six distinct model setups divided into two structural paradigms:

1) *Group A (End-to-End Models)*: A1 (SARIMA): Seasonal Autoregressive Integrated Moving Average based on the classical mathematical foundation of Box and Jenkins [1], operating directly on the raw, un-decomposed target.

A2 (XGBoost): A Gradient Boosted Decision Tree architecture formulated by Chen and Guestrin [2], transforming a dense temporal feature matrix directly to Y_{t+h} .

A3 (LSTM): A recurrent deep learning network composed of gated memory blocks designed by Hochreiter and Schmidhuber [6] to internally capture sequential context from Y_t .

2) *Group B (Decompose-then-Forecast Models)*: The Group B pipelines implement a decoupled strategy:

$$\hat{Y}_{t+h} = \hat{T}_{t+h} + S_{\text{pattern}}[\text{doy}(t+h)] + \hat{R}_{t+h} \quad (5)$$

where S_{pattern} is an extracted deterministic daily seasonal pattern indexed by day-of-year ($\text{doy} \in [1, 366]$).

B1 (STL+SARIMA): The combined component ($T_t + R_t$) is modeled via an automated SARIMA structure.

B2 (STL+XGBoost): The high-frequency remainder R_t is forecasted by XGBoost [2], while the trend is modeled via a local linear estimator.

B3 (STL+LSTM): The network [6] is trained exclusively on the bounded sequence of the remainder R_t .

III. EXPERIMENTAL CONFIGURATION

A. Data Characterization and Data Splitting

The empirical data consists of daily historical records from the Hanoi weather station (21.03°N, 105.85°E) officially curated by the Visual Crossing Corporation database [12], spanning Jan 1, 2015 to Mar 31, 2025 ($n = 3,742$). The primary target is daily mean temperature (temp, °C).

To ensure realistic evaluation free from temporal data leakage, a strict chronological data split is applied. The Training Set spans from 2015-01-01 to 2022-12-31 ($n_{\text{train}} = 2,922$). The Validation Set covers 2023-01-01 to 2023-12-31 ($n_{\text{val}} = 365$). The Test Set encompasses 2024-01-01 to 2024-12-31 ($n_{\text{test}} = 365$).

B. Hyperparameter Optimization and Setup

The engineering configuration mirrors the exact execution architecture defined in the centralized pipeline.

For the STL Specs, the seasonal period is set to $M = 365$ [3]. Robust Loess optimization is active with linear degrees enforced for both trend and seasonal tracking ($\text{trend_deg} = 1$, $\text{seasonal_deg} = 1$).

For the SARIMA Optimization, automated order selection utilizes heuristics implemented within the standard algorithmic framework [7], constrained to $\max(p, q) = 3$ and seasonal $\max(P, Q) = 2$. To maintain scalability over multi-step updates, full structural refitting is restricted to a 30-day sliding batch window, with intermittent updates handled recursively.

For the XGBoost Architecture, the model is configured with $n_{\text{estimators}} = 500$, $\text{max_depth} = 6$, and $\text{learning_rate} = 0.05$ [2]. The input feature space includes temporal lag days $L \in \{1, 2, 3, 7, 14, 30, 365\}$ and rolling statistics calculated across window lengths of 7 and 30 days.

For the LSTM Specifications, the network is configured as a stacked recurrent architecture [6] with dense layers containing [64, 32] hidden units, standard dropout of 0.2, and a lookback window fixed at 30 days. Optimization is driven by Adam with early stopping triggered after 10 epochs of validation loss stagnation.

IV. EMPIRICAL RESULTS AND VISUALIZATION

A. Structural Diagnostic Findings

Applying the mathematical variance criteria proposed by Wang et al. [11] to the training partitions yields highly skewed results regarding the data's composition. The Strength of Trend (F_T) is computed as 0.0140, while the Strength of Seasonality (F_S) is evaluated at 0.8016. The baseline data possesses virtually zero linear trend information ($F_T \approx 1.4\%$), whereas the annual seasonal envelope controls over 80.16% of total sample variance.

To evaluate the mathematical properties prior to model execution, standard non-stationarity diagnostics are implemented. The Augmented Dickey-Fuller (ADF) test [4] strongly rejects the unit root null hypothesis ($p < 0.001$), while the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test [8] fails to reject the stationarity assumption around a deterministic seasonal mean ($p > 0.1$). Furthermore, a Ljung-Box test [9] on the isolated remainder component rejects the white noise null hypothesis up to lag 20 ($Q^* = 2568.38, p < 0.001$), proving that significant complex non-linear information remains embedded within the remainder sequence.

B. Symmetrical Model Evaluation and Degradation Plots

Table I summarizes the cross-model error metrics evaluated systematically across horizons $t+1$, $t+3$, and $t+7$.

The end-to-end LSTM model consistently demonstrates superior performance across all evaluated forecasting horizons. Crucially, the empirical data yields a uniform negative result regarding the deployment of STL preprocessing: every hybrid model variation in Group B exhibits strict performance degradation compared to its Group A equivalent. At horizon $t+1$,

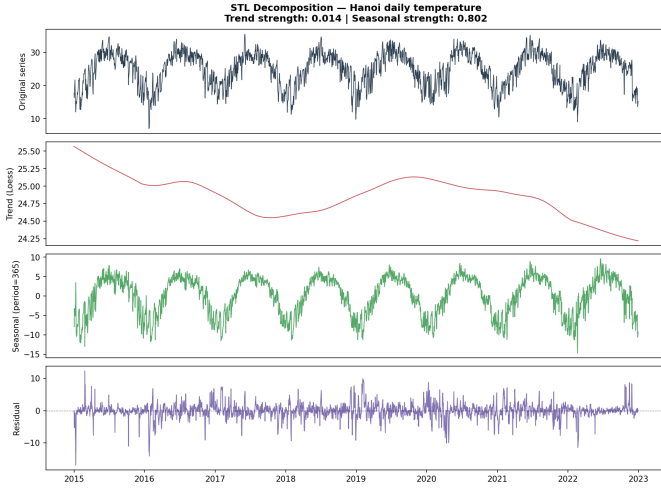


Fig. 1: Visual diagnostics of STL decomposition. The red line (Trend) demonstrates a nearly flat profile ($F_T = 0.014$), confirming that explicit trend modeling injects additive noise into the forecasting pipeline rather than capturing a meaningful signal.

TABLE I: Comprehensive Forecasting Performance Matrix

Horizon	Model	RMSE ($^{\circ}\text{C}$)	MAE ($^{\circ}\text{C}$)	MAPE (%)	Skill
t+1	SARIMA	1.7199	1.3251	5.8025	-0.0721
	XGBoost	1.6005	1.2081	5.2338	0.0023
	LSTM (Best)	1.5450	1.2032	5.3205	0.0369
	STL+SARIMA	2.0538	1.5524	7.0039	-0.2802
	STL+XGBoost	2.8000	2.1269	9.6795	-0.7453
	STL+LSTM	2.8482	2.1055	9.7284	-0.7754
t+3	SARIMA	2.3989	1.8131	8.2410	0.1899
	XGBoost	2.9678	2.2254	10.1365	-0.0022
	LSTM (Best)	2.1419	1.6464	7.5381	0.2767
	STL+SARIMA	2.9569	2.1397	9.9966	0.0015
	STL+XGBoost	2.8004	2.1278	9.6846	0.0543
	STL+LSTM	2.9811	2.1970	10.1634	-0.0067
t+7	SARIMA	2.7136	2.0824	9.5897	0.2651
	XGBoost	3.7266	2.8079	13.1039	-0.0093
	LSTM (Best)	2.5316	1.9289	9.0248	0.3144
	STL+SARIMA	3.3000	2.3445	11.1895	0.1063
	STL+XGBoost	2.8006	2.1280	9.6849	0.2415
	STL+LSTM	3.1118	2.2856	10.5996	0.1572

extracting the STL components causes the LSTM error profile to jump from 1.5450°C to 2.8482°C .

The predictive accuracy drops severely when the data undergoes explicit decomposition. To verify that this degradation is not an artifact of random sample variation, a Diebold-Mariano predictive accuracy test [5] was conducted. The test confirms that the accuracy difference between LSTM and STL+LSTM is highly significant, particularly at the immediate $t + 1$ step ($DM = -5.44, p < 0.001$).

V. DISCUSSION AND THEORETICAL BREAKDOWN

The consistent failure of the STL hybrid models across all configurations contradicts the common assumption that

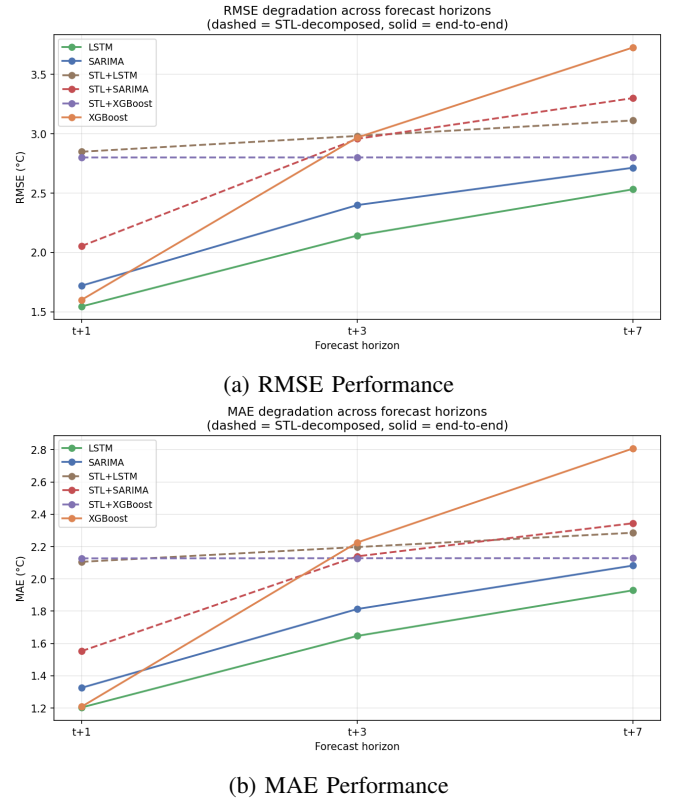


Fig. 2: Forecasting error degradation across forecast horizons for Group A and Group B models.

decomposition simplifies the forecasting problem. This systematic drop in performance can be explained by three distinct structural mechanisms:

1. *The Trend-Induced Noise Injection Mechanism:* Because the empirical strength of trend is nearly zero ($F_T = 0.0140$), the local Loess tracker captures random multi-year fluctuations rather than a persistent underlying signal. When this unstable trend is extrapolated forward, it acts as additive noise, compounding errors during the final profile reconstruction.

2. *Context Stripping and Residual Overfitting:* Machine learning models like XGBoost and deep neural networks like LSTM excel at learning complex seasonal shapes and temporal interactions directly from unadjusted data sequences. Isolating the seasonal pattern removes this essential context. Forcing these architectures to optimize exclusively on highly volatile remainder data (R_t) strips away the smooth seasonal boundaries, causing the models to overfit to short-term noise.

3. *Multi-Step Error Propagation:* In the end-to-end setups, errors across extended horizons ($t+3 \rightarrow t+7$) increase gracefully. In contrast, the hybrid configurations suffer from compounding errors across three independent forecasting channels ($\hat{T}$, S , $\hat{R}$). This multi-component compounding degrades accuracy faster than direct modeling, as evidenced by the high MAE metrics.

Based on these findings, we establish a definitive architectural guardrail: explicit decomposition preprocessing should

be restricted to datasets demonstrating a clear, strong trend component ($F_T > 0.30$). When $F_T \rightarrow 0$, end-to-end recursive structures should be preferred.

VI. CONCLUSION

This study provides a rigorous empirical evaluation of the structural limits of decomposition-based forecasting. Using an extensive meteorological dataset from Hanoi, we demonstrated that applying STL decomposition to a time series with a weak trend ($F_T = 0.0140$) systematically degrades predictive performance across statistical, machine learning, and deep learning models. The end-to-end LSTM architecture achieved the highest accuracy across all horizons, outperforming the preprocessed alternative by avoiding the noise injection and overfitting inherent in the hybrid pipeline. Future research will focus on dynamically activating decomposition layers based on real-time variance calculations over rolling windows.

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